HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Cleaned, structured, and validated survey and panel datasets, including recoding variables and merging files, for over **50 quantitative research projects**.
- Developed and implemented statistical analysis plans, producing **descriptive statistics**, cross-tabulations, and significance tests using SPSS.
- Translated complex statistical results into clear tables, charts, and findings for **non-specialist research colleagues**, supporting client recommendations.
- Applied weighting strategies and analyzed changes across study phases for **longitudinal research datasets**, ensuring methodological consistency.
- Supported researchers with questionnaire design by evaluating analytical feasibility and ensuring **methodological consistency**.
- Maintained accurate documentation and traceability for data processing pipelines, contributing to **improved analytical workflows** and quality control.
- Automated data preparation tasks and report generation using **Python scripting**, reducing manual processing time by an estimated **30%**.

SKILLS

Statistical Software – SPSS, R, RStudio

Data Analysis & Visualization – Power BI, Tableau, Excel (PivotTables, complex formulas), DAX, Power Query

Survey Methodology – Survey Data Analysis, Panel Data Analysis, Questionnaire Design, Sampling, Weighting, Bias, Statistical Significance, Multivariate Analysis, Segmentation, Regression, Longitudinal Research

Programming & Scripting – Python, SQL

Data Management – Data Cleaning, Data Structuring, Data Validation, Merging Files, Identifying Missing Values, Duplicate Detection, Inconsistencies

Reporting & Documentation – Research Findings, Tables, Charts, Documentation, Traceability

PROJECTS

Survey Data Analysis for Health Research

- Cleaned and structured a **panel dataset of 15,000+ respondents** for a multi-phase health study, identifying and handling missing values and duplicates.
- Performed statistical analysis using **SPSS**, including cross-tabulations and t-tests, to compare respondent groups across demographic segments.
- Generated visualisations and summary tables in **Power BI** to communicate key findings on health behaviours to the research team.

Automated Reporting Pipeline with Python

- Developed Python scripts to automate the extraction, transformation, and loading (ETL) of **survey response data** from multiple sources into a centralized repository.
- Implemented data validation checks using **pandas** to ensure accuracy and consistency before loading, catching approximately **85%** of data integrity issues.
- Generated automated weekly summary reports, reducing manual reporting effort by **4 hours per week**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)